

Paper II v1.2 Internal Audit Final Report

Internal author-side audit — not external

21 August 2026 — Overall verdict: PASS

Protocol: INTERNAL_AUDIT_STANDARD_v1.3.

Lean execution: recorded main and supplementary builds reviewed; full builds not rerun.

1 Frozen target

English SHA-256: 7215e14bbea8ab2bf208dcdd1efa050cd2b72c997eee2efe504a1e6817c68882.
Spanish SHA-256: d0d1df05eb267a51db2ccc100dd9725dcde9b03dbb95c8a730742e357eb0f4dc.
Archive SHA-256: ee2d05cc40d943ca92f8f7bf3e5dd83c2692518ddea5e2ca4f7686ccb1ac3895.
Recorded builds: 8,061 main jobs and 8,032 supplementary jobs.

2 Gate results

Gate	Verdict	Evidence
G0	PASS	57 static checks; exact archive, manifests and six artifacts.
G1	PASS	Theorem, branches, degeneracies, scope and quantifiers.
G2	PASS after harness correction	195 LPs; $n \leq 5000$; 139 chordal graphs; 931 copy pairs.
G3	PASS	Main theorem and auxiliary surfaces matched.
G4	PASS recorded build	Both logs and 16 axiom surfaces reviewed.
G5	PASS	English/Spanish hierarchy, figures and identifiers aligned.
G6	PASS	23/24 pages; fonts embedded; every page inspected.
G7	PASS internal	Scoped negative literature result; external specialist gate open.
G8	PASS	Self-contained v1.2 evidence perimeter.

3 Corrections

Stale formal-package references were replaced by the delivered v1.2 archive, hash and both actual build logs. All manuscript formats and hashes were regenerated. No theorem statement, extremal value or proof step changed.

4 Overall verdict

PASS. No unresolved internal blocker or major finding remains. Independent reconstruction and all external review/release gates remain open.